

Surface Characterization

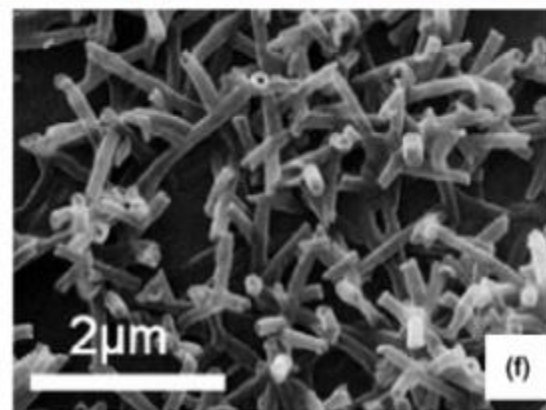
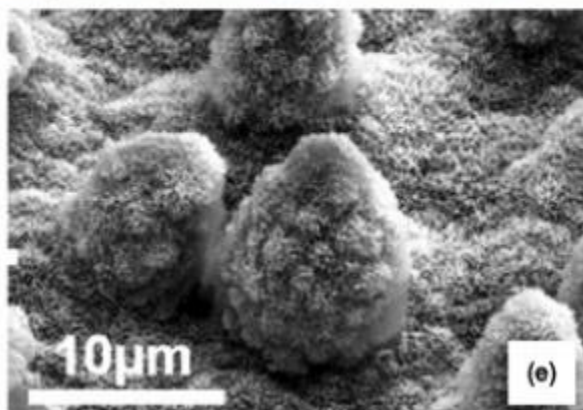
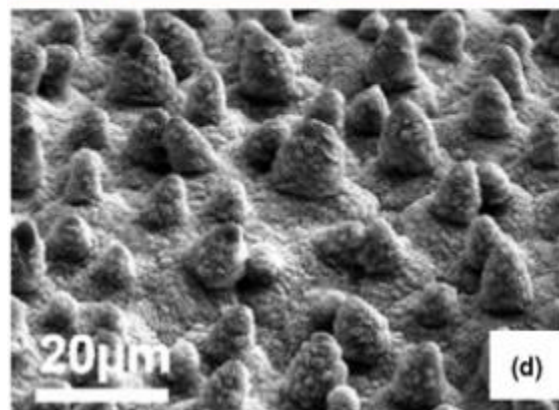
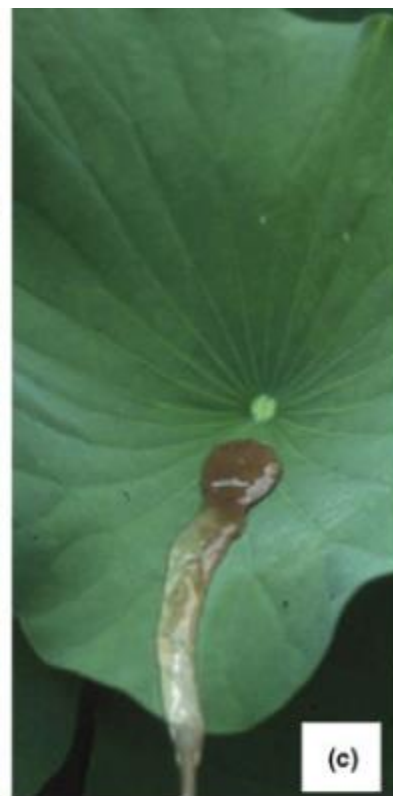
Contact Angle Analysis

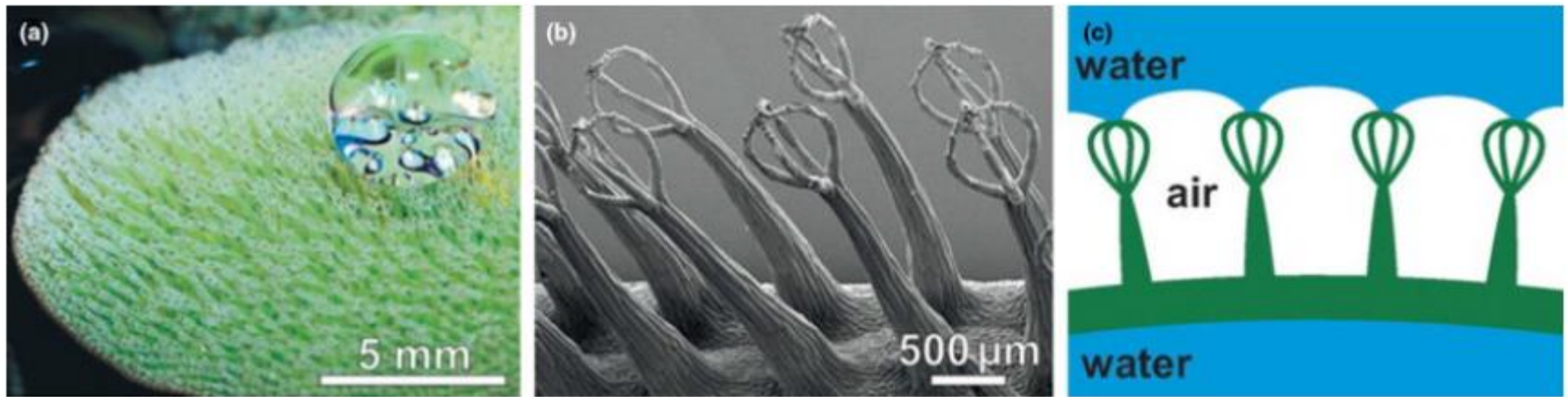
Contents

- Hydrophilic (亲水性) and hydrophobic (疏水性) in nature
- **Contact Angle and Wettability**
- **Young's equation**
- **Sessile drop method (固着液滴法)**
- Wetting of rough surfaces
- Wetting of heterogeneous surfaces



This butterfly wing has structures which split a water droplet as it bounces off



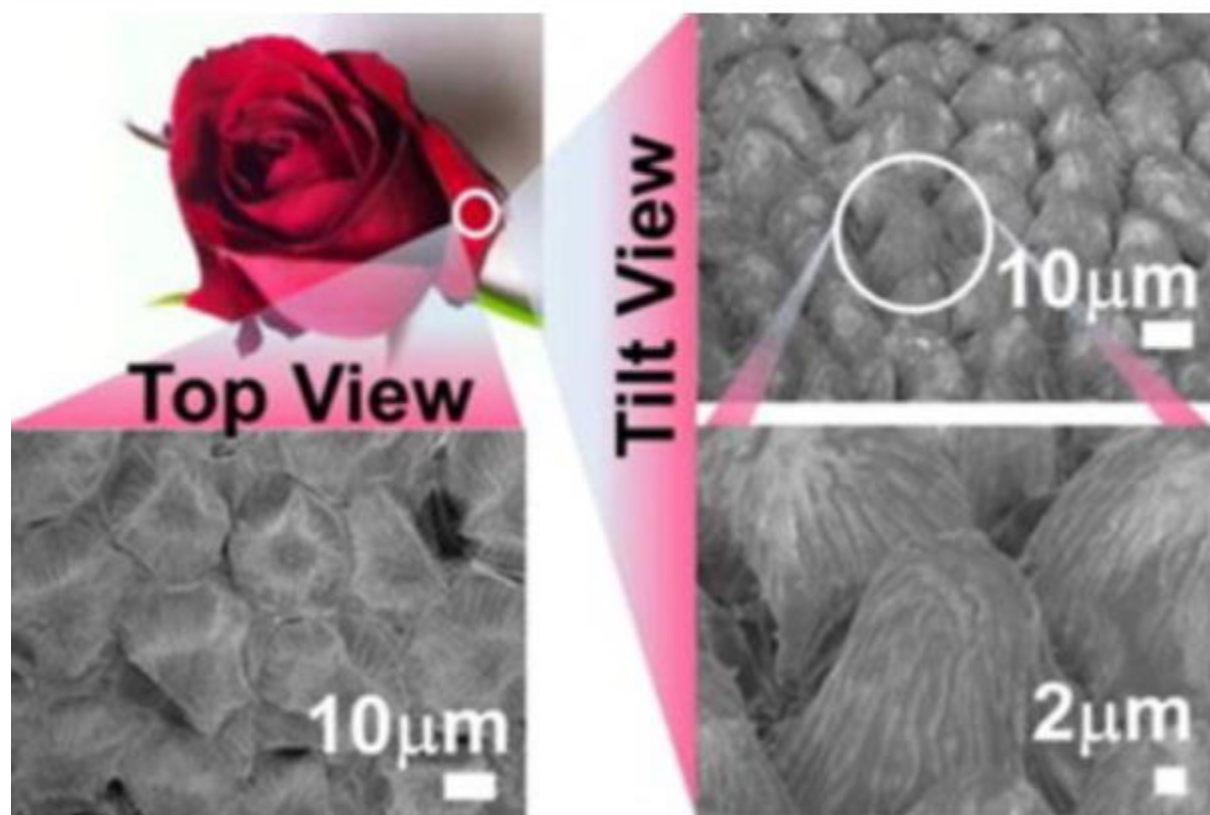


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鼠尾草

Figure 3. Images of a superhydrophobic *Salvinia molesta* at different magnifications. [19], Copyright 2010. Reprinted with permission from Wiley-VCH, Germany.

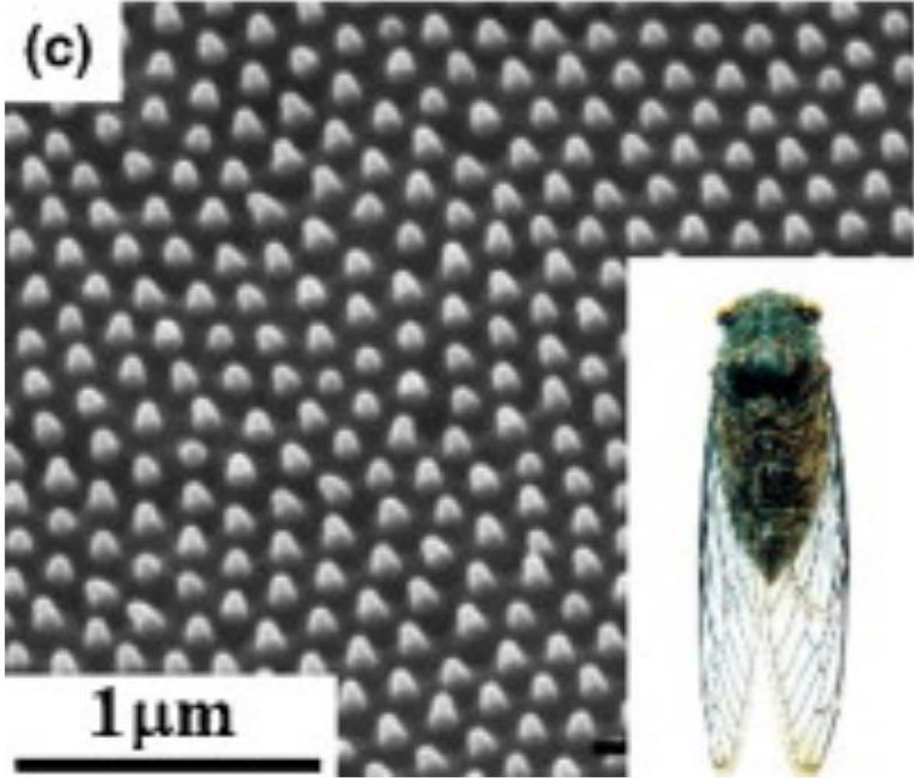


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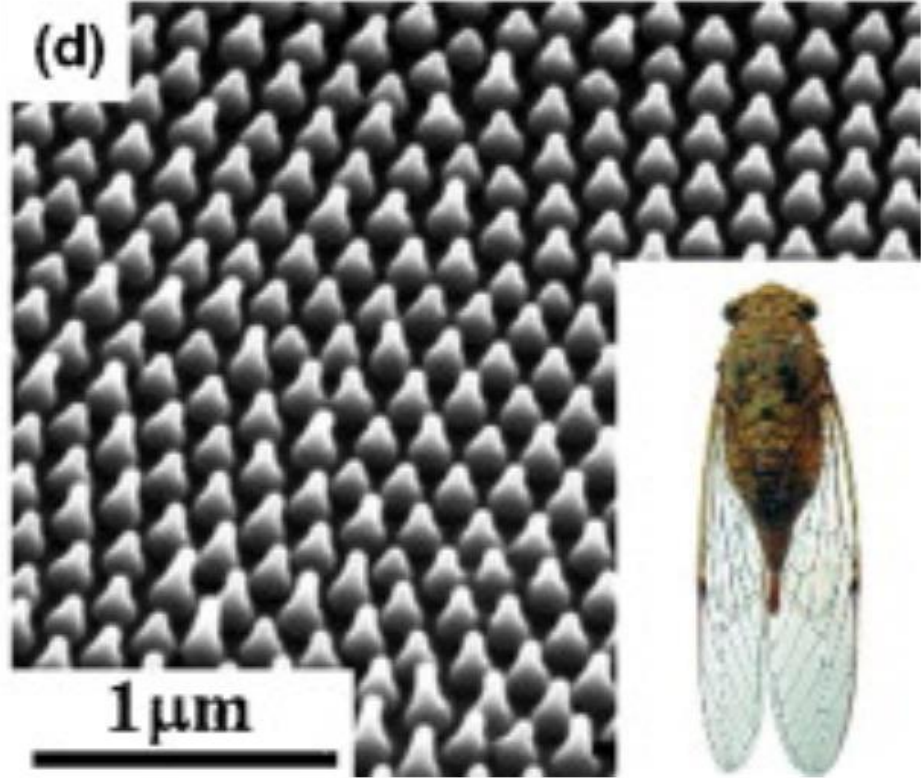
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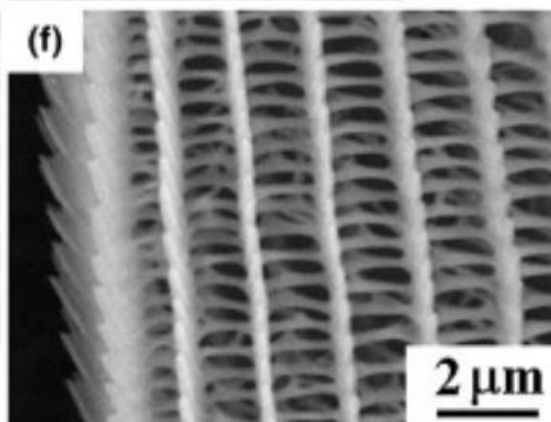
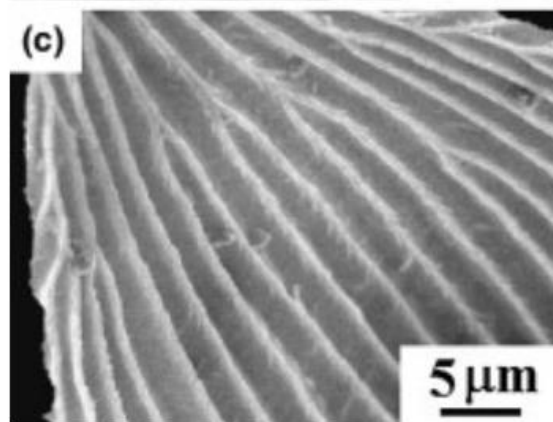
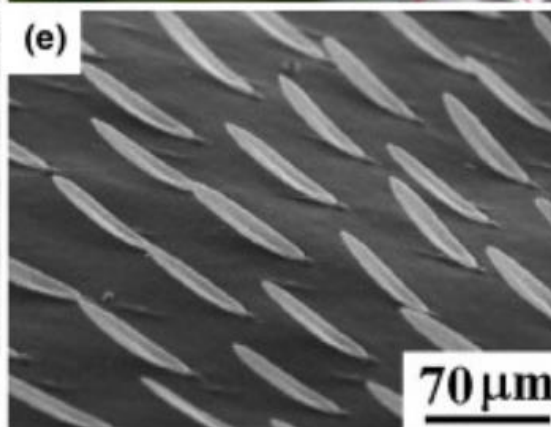
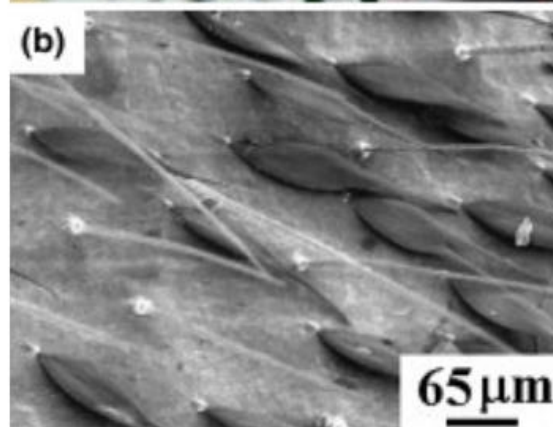
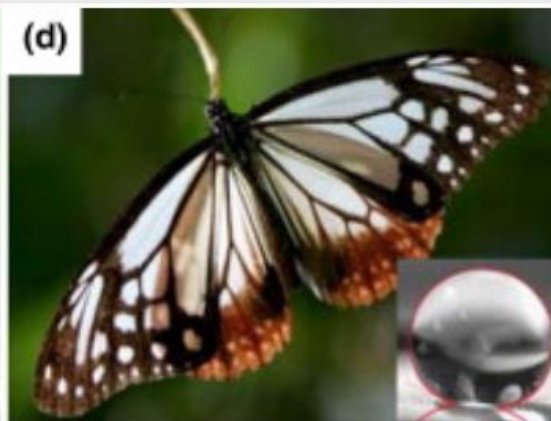
Figure 4. Images of a superhydrophobic red rose petal with high adhesion at different magnifications. [36], Copyright 2014. Reprinted with permission from the American Chemical Society, USA.

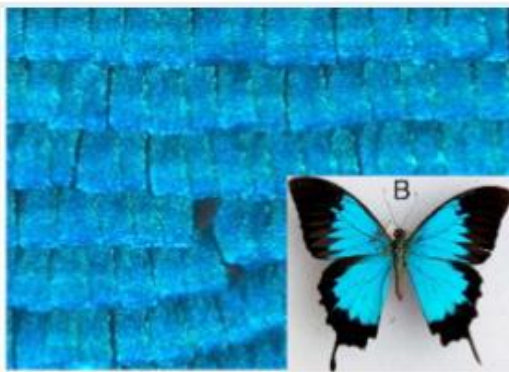
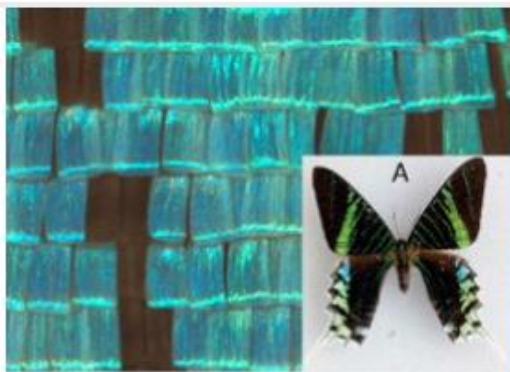
(c)



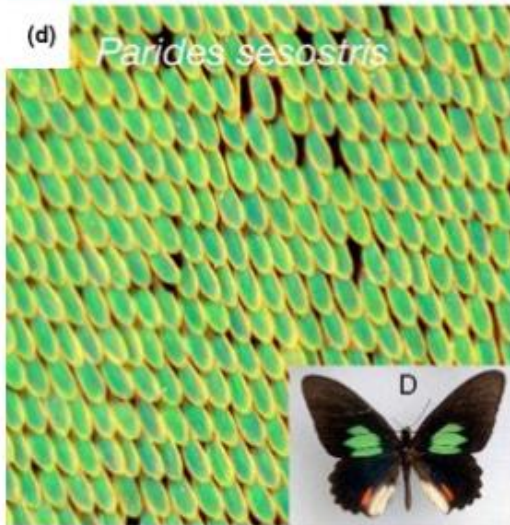
(d)



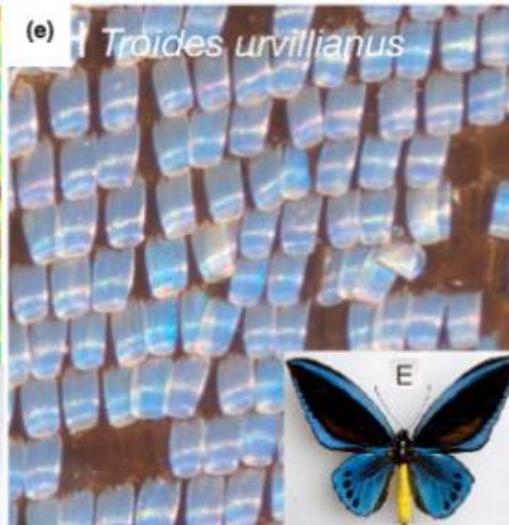




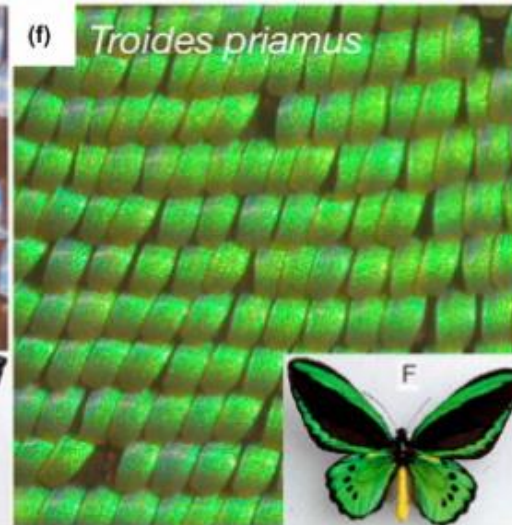
(d) *Parides sesostris*



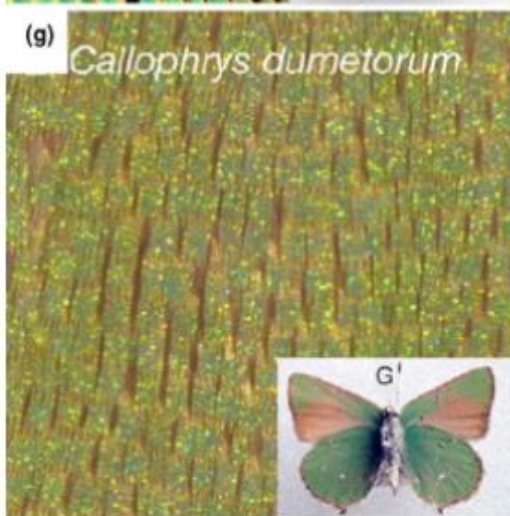
(e) *Troides urvillianus*



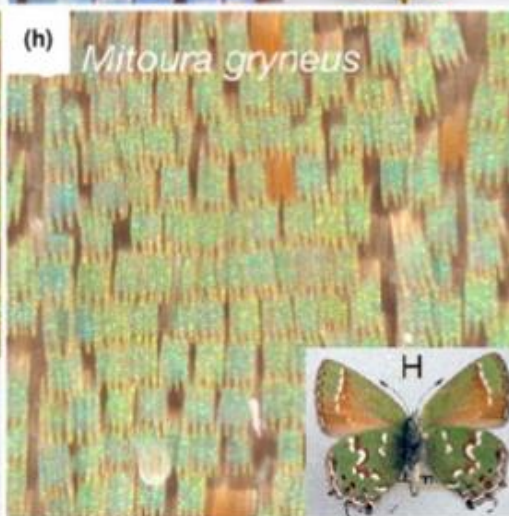
(f) *Troides priamus*



(g) *Callophrys dumetorum*

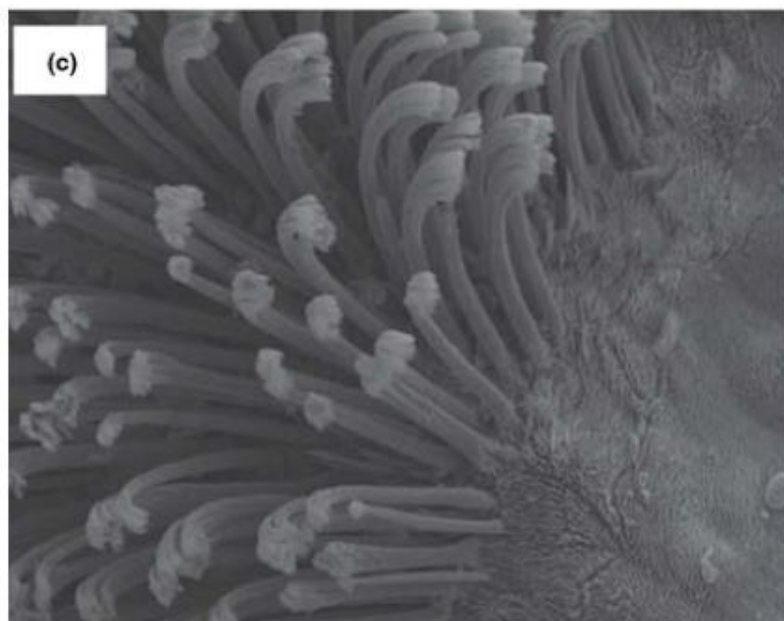
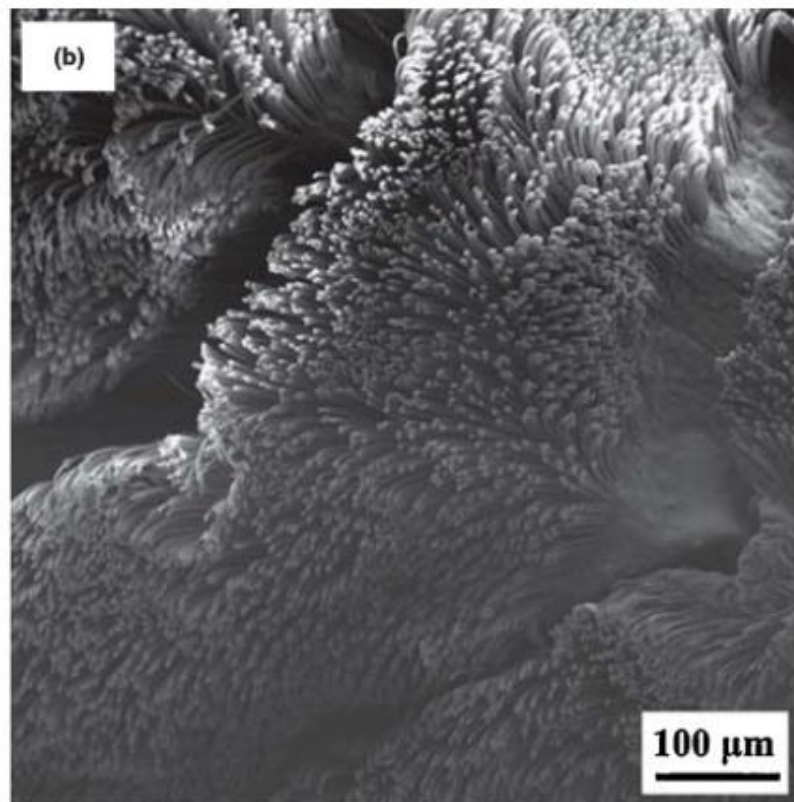


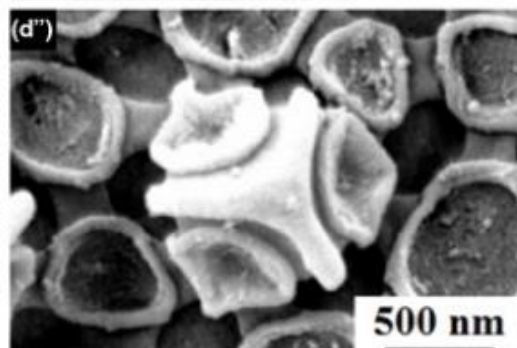
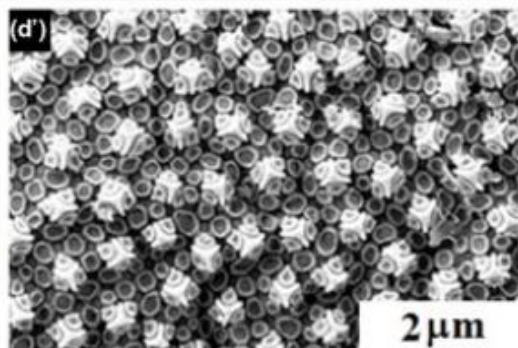
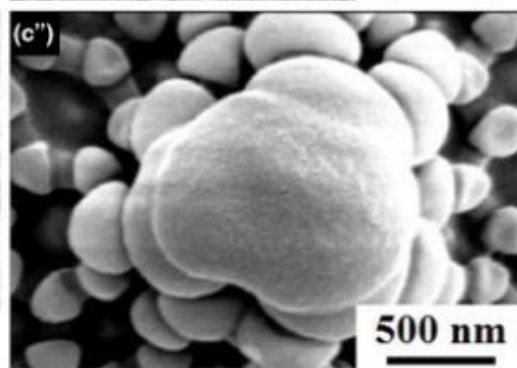
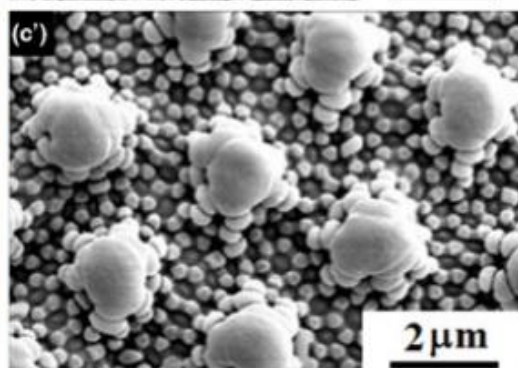
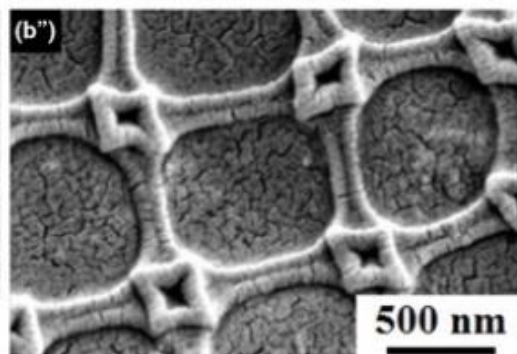
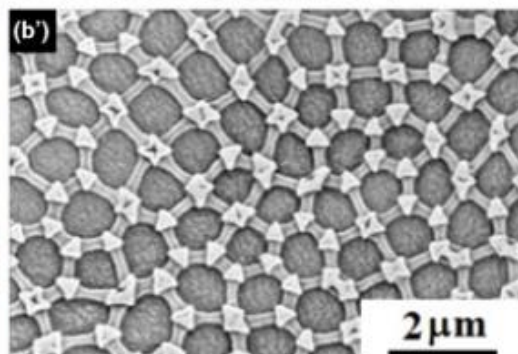
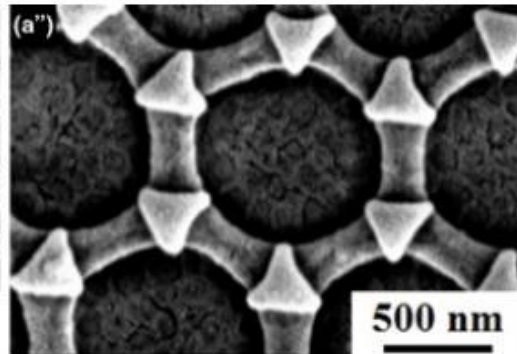
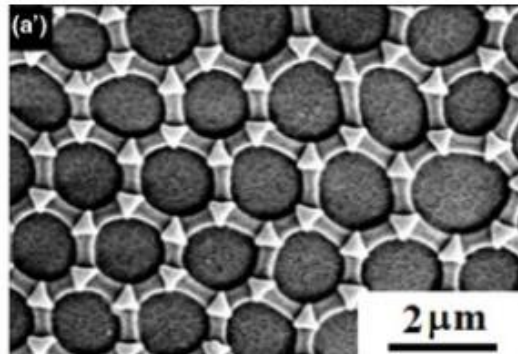
(h) *Mitoura gryneus*



(i) *Morpho aega*





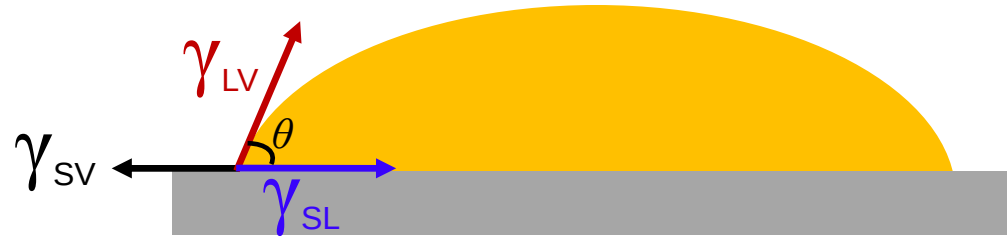


Contact Angle and Wettability

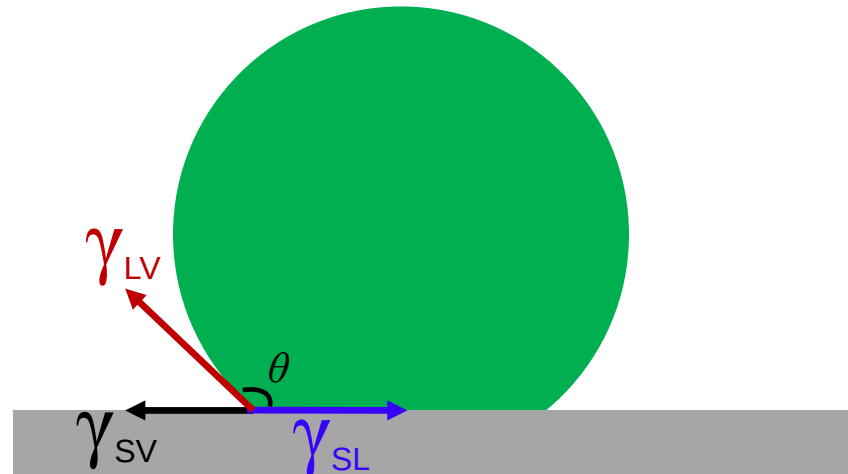
Contact Angle and Wettability

Contact Angle is an angle formed by a liquid at the three-phase boundary where a liquid, gas and solid interest.

Case 1:



Case 2:

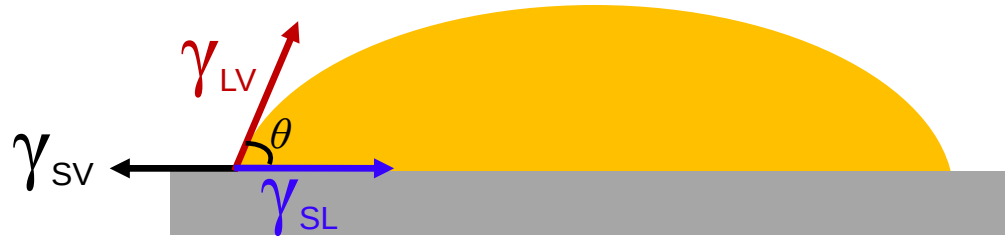


γ_{SV} : is the solid/vapor surface tension

γ_{SL} : is the solid/liquid surface tension

is the liquid/vapor surface tension

θ is the contact angle



$$\gamma_{SV} - \gamma_{SL} - \gamma_{LV} \cos\theta = 0$$

(Young equation)

Only two measurable quantities:

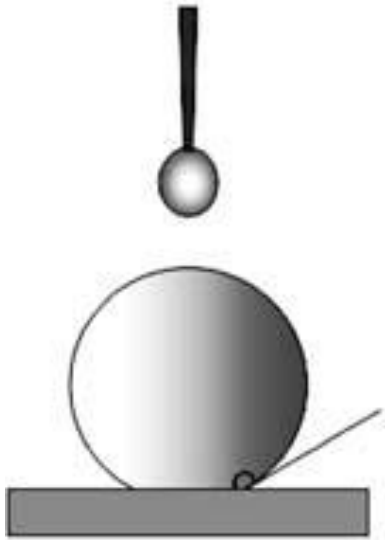
1. the contact angle θ ;
2. the liquid-vapor surface tension, γ_{LV} .

- θ , can be determined during the test
- γ_{LV} can be determined and calculated using the pendant drop method (吊坠法) by fitting the shape of the suspended droplet (in a captured video image) to the Young–Laplace equation

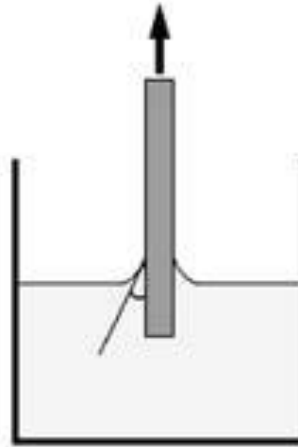
To have a precise measurement of contact angle, a few assumptions have to be made:

1. All approaches rely on the validity and applicability of Young's equation for surface energetics from experimental contact angles.
2. Only pure liquids are used; surfactant solutions or mixtures of liquids should not be used as preferential adsorption can occur.
3. The values of γ_{lv} , γ_{sv} , and γ_{sl} are assumed to be constant during the experiment, that is, there should be no physical or chemical reaction between the solid and the liquid.
4. The test liquid surface tensions should be higher than the anticipated solid surface energy.
5. The values of γ_{sv} going from liquid to liquid are also assumed to be constant, that is, independent of the liquids used.

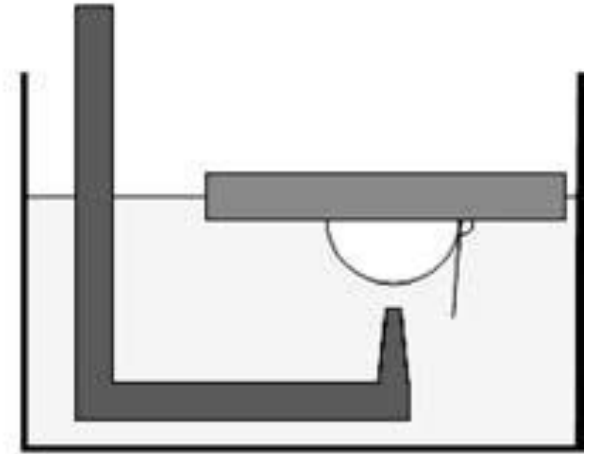
Five ways that the contact angle (θ) can be measured



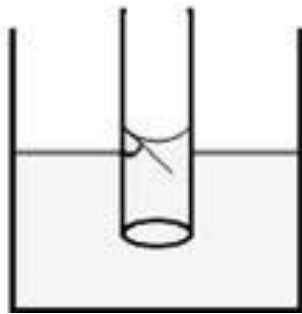
(a) Sessile drop method



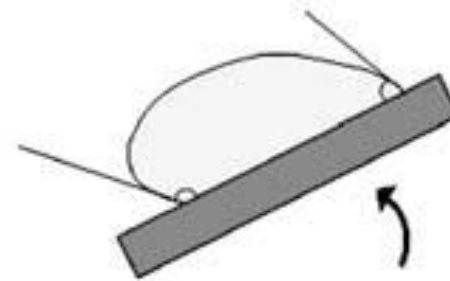
(b) Wilhelmy plate method



(c) Captive air bubble method

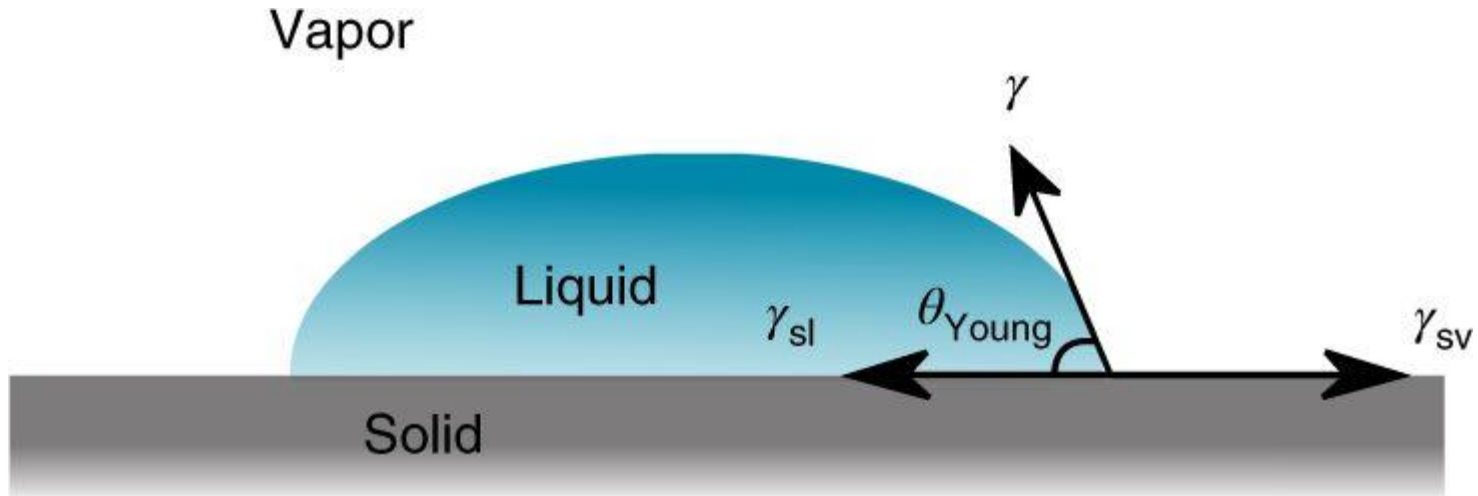


(d) Capillary rise method



(e) Tilting substrate method

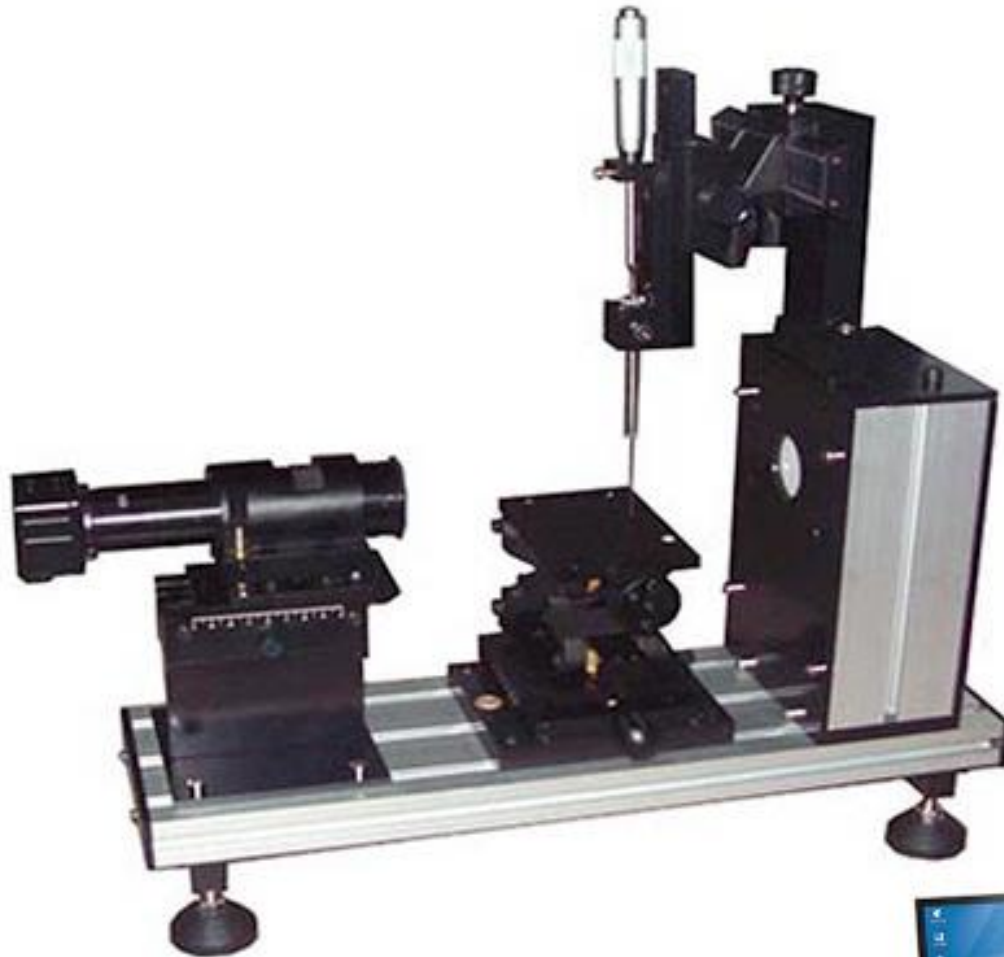
Sessile drop method



- The contact angle is measured by a contact angle goniometer
- An optical system to capture the profile of the liquid droplet on a solid substrate
- The angle formed between the liquid–solid interface and the liquid–vapor interface is the contact angle
- A typical system uses a high-resolution camera and software to capture and analyze the contact angle

Sessile drop method-goniometer

High-speed
camera



Goniometer

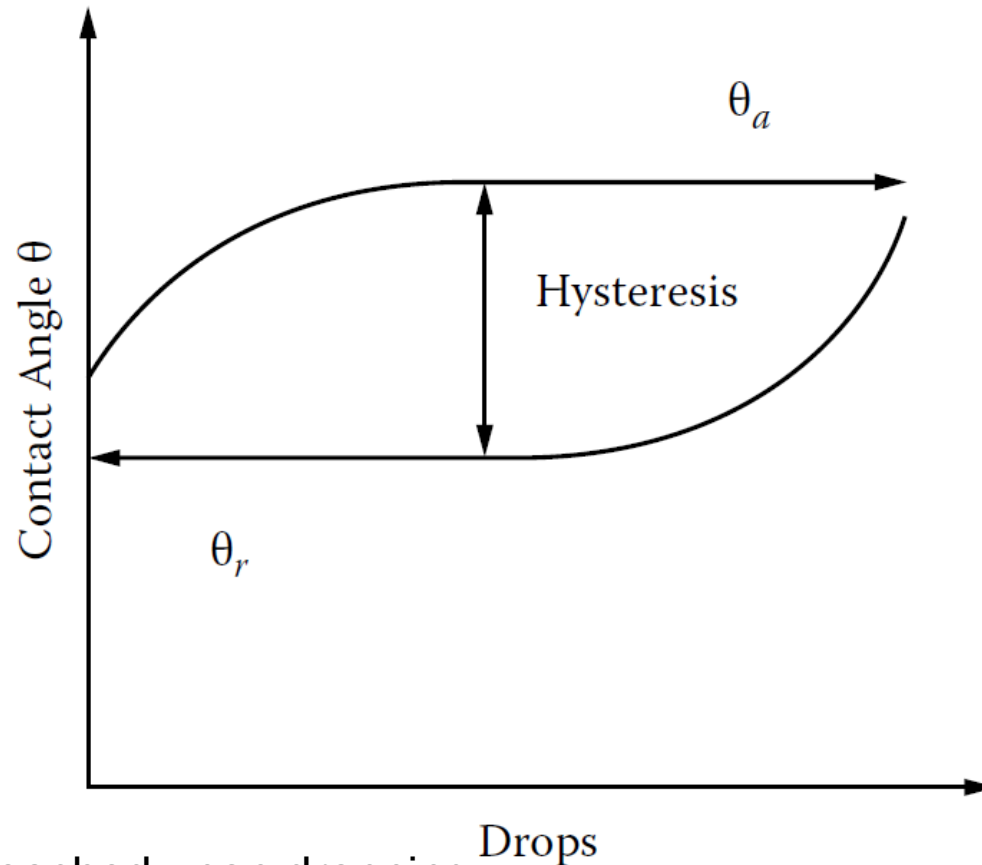


Sessile drop method { Static
Dynamic

Static, interactions may change the contact angle:

- Evaporation of the liquid
 - Migration of surfactants from the solid surface to the liquid surface
 - Substances dissolved in the drop migrating to the surface (or in the opposite direction)
 - Chemical reactions between the solid and liquid
 - The solid being dissolved or swollen by the liquid
- **Advantage: the needle does not remain in the drop during the measurement**

Dynamic:



- A plateau is reached upon dropping
- The advancing and retreating angles are usually not equal
- The hysteresis (滞后) can result from sample prehydration, surface roughness, chemical heterogeneity, evaporation, or molecular movement.
- The receding contact angle allows one to measure the degree of hysteresis inherent in the sample surface

Tip: The flow rate should not be too high

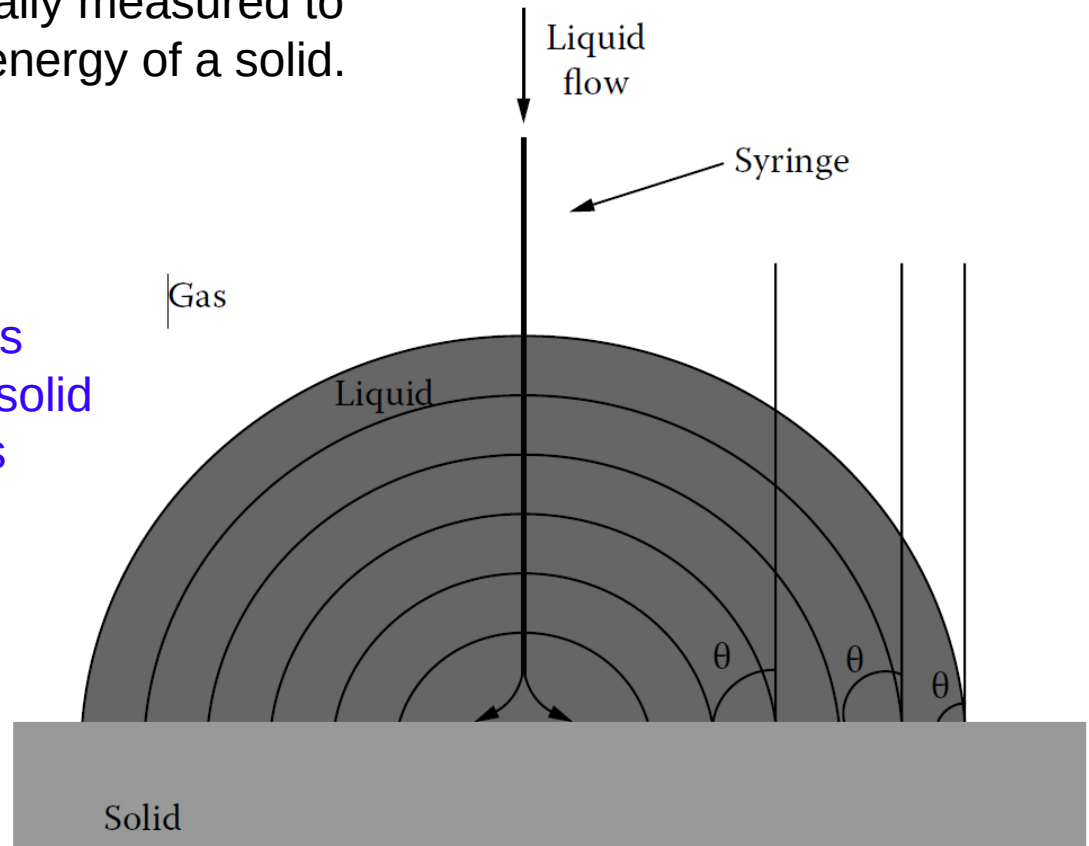
Dynamic

Advancing Angle

- Advancing angles simulate a fresh surface for the contact angle
- Most reproducible way of measuring contact angles
- Advancing angles are normally measured to determine the surface free energy of a solid.

Receding Angle

- Possible to make statements about the roughness of the solid or chemical inhomogeneties
- Not suitable for calculating surface energies



Note that the contact angle is always denoted by symbol θ .

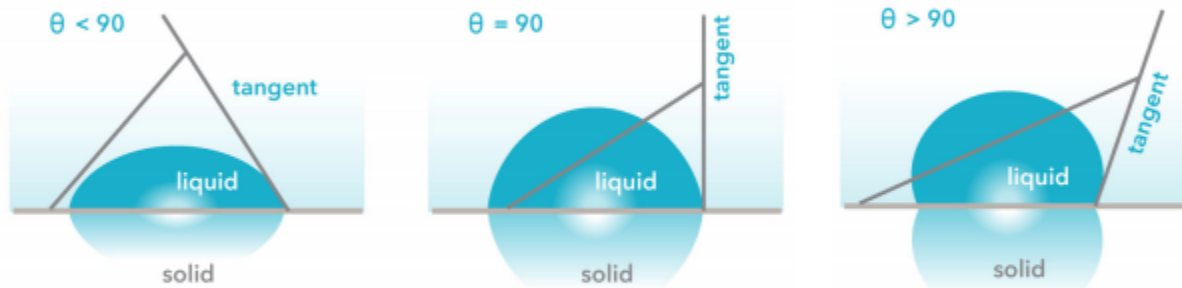
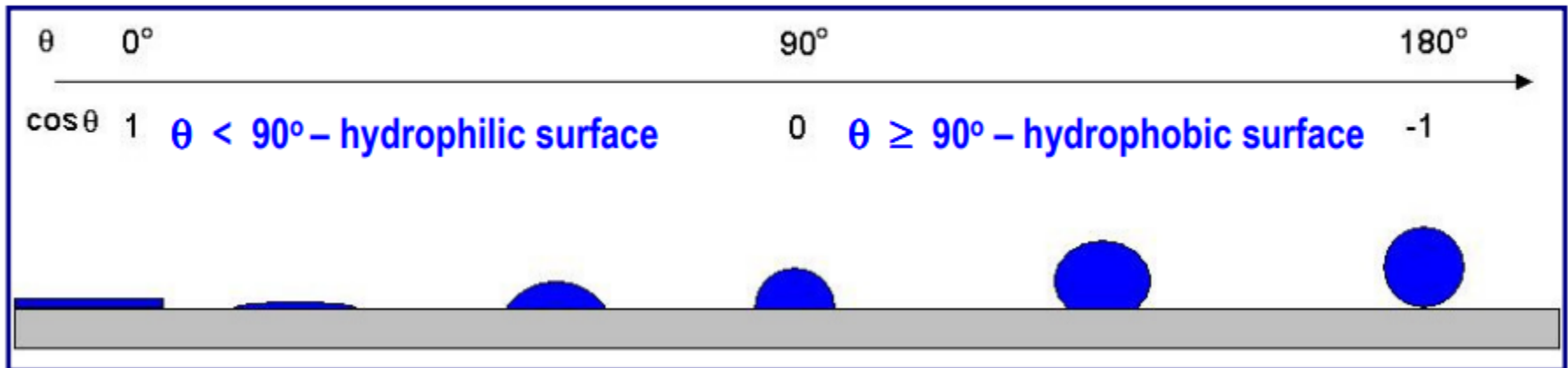
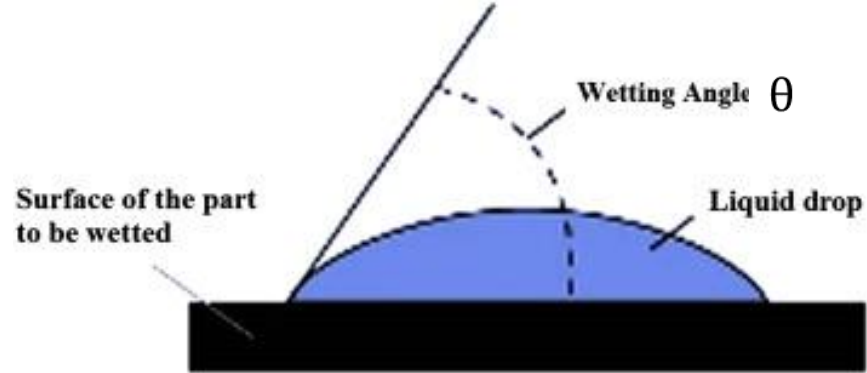
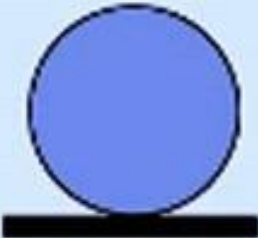


Fig. 13.1. Scheme of contact angles on the hydrophobic and hydrophilic surfaces

<http://www.attension.com/?id=1092&cid=>



$\theta = 0^\circ$		Spreading
$\theta < 90^\circ$		Good Wetting
$\theta = 90^\circ$		Incomplete wetting
$\theta > 90^\circ$		Incomplete wetting
$\theta = 180^\circ$		Nonwetting

$\theta > 150^\circ$,
superhydrophobic

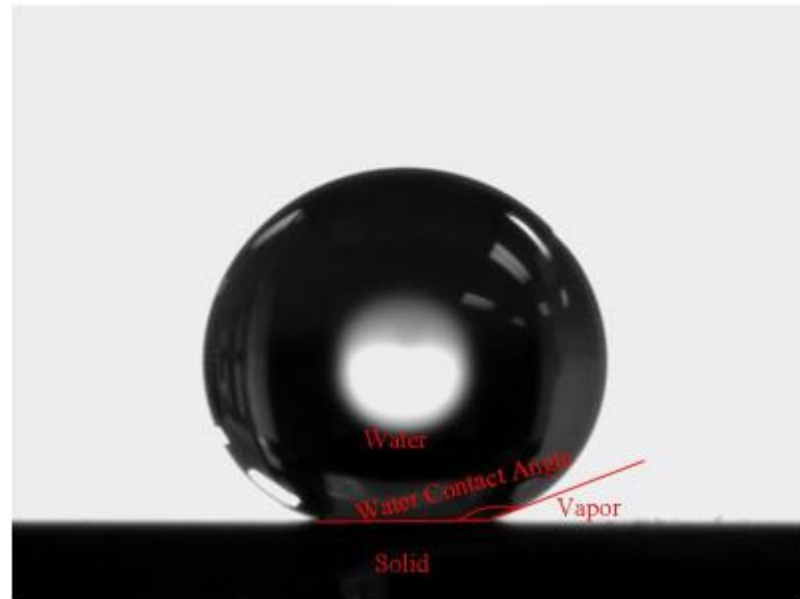
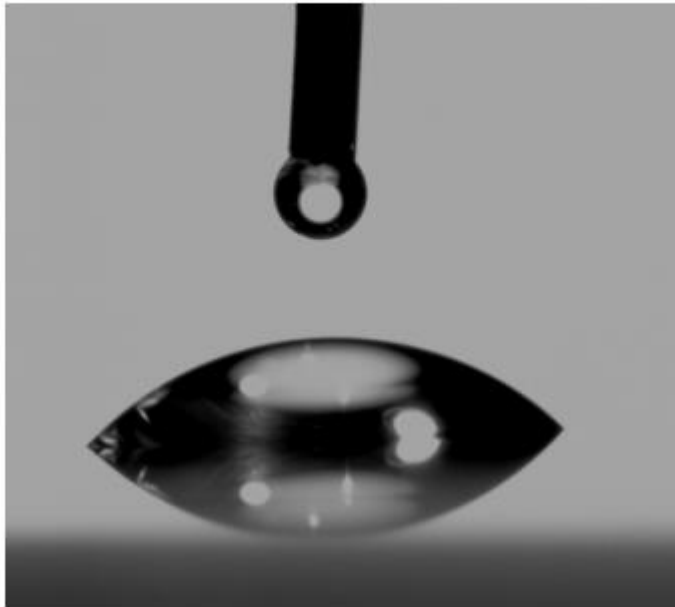


Fig. 13.2. Photographs of a water droplet on the hydrophilic and hydrophobic surfaces

[http://en.wikipedia.org/wiki/
Contact_angle#Typical_contact_angles](http://en.wikipedia.org/wiki/Contact_angle#Typical_contact_angles)

[http://www.google.pl/imgres?imgurl=http://
dailyheadlines.uark.edu/ images/water
contact_angle.jpg&imgrefurl= http://
dailyheadlines.uark.edu/](http://www.google.pl/imgres?imgurl=http://dailyheadlines.uark.edu/images/water_contact_angle.jpg&imgrefurl=http://dailyheadlines.uark.edu/)

A)



B)

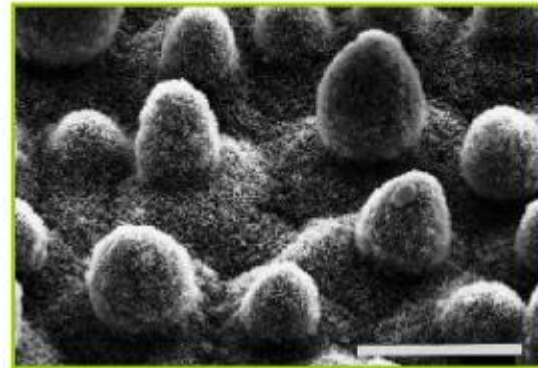


Fig. 13.10.

A) A water droplet on a lotus leaf.

(From: <http://www.botanik.unibonn.de/system/lotus/en/prinzip.html.html>).

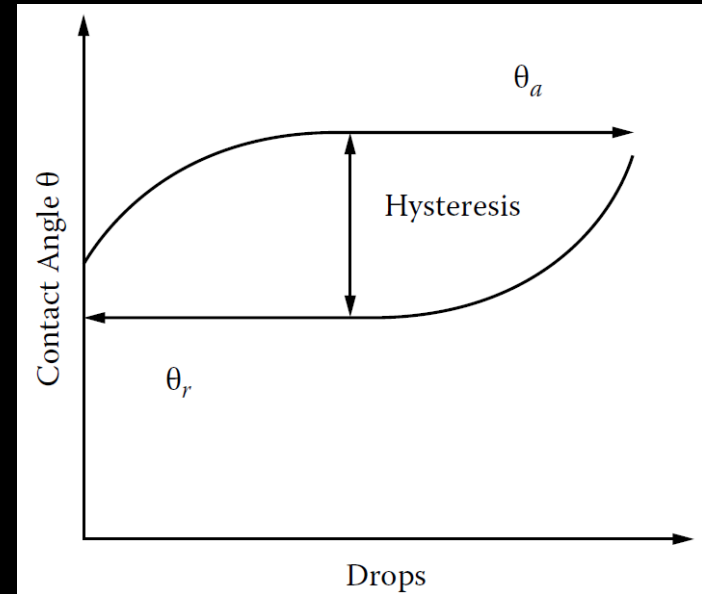
B) SEM-image of lotus leaf. The micro structural epidermal cells are covered with nanoscopic wax crystals. Bar: 20 μm .

(from W. Barthlott and C. Neinhuis. *Planta* 202. 1(1997).

The air trapped between the wax roughness on the leaf surface minimizes the contact area of the water droplet. There are several naturally superhydrophobic surfaces, and the most known is that of lotus leaf on which θ equals up to 170° .

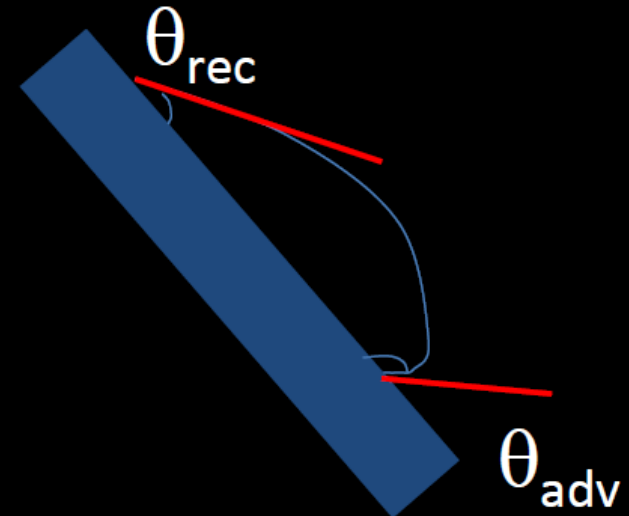
There are different contact angles...

- Three different contact angles can be defined, the static, receding, and the advancing contact angle



- For a real surface

$$\theta_{\text{rec}} < \theta_{\text{stat}} < \theta_{\text{adv}}$$



Two cases:

Wenzel equation

- described wetting of a rough, hydrophilic and one-component film

Cassie equation

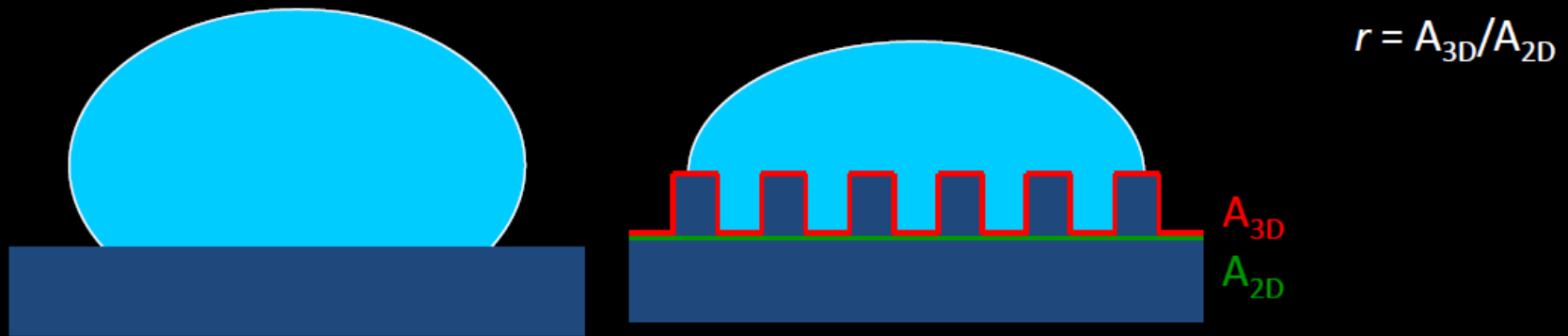
- described wetting of a flat, chemically heterogeneous film

Wetting of rough surfaces: the Wenzel equation

- The wetting of a rough, hydrophilic, one-component surface is often described by the Wenzel equation, where the apparent contact angle is given by

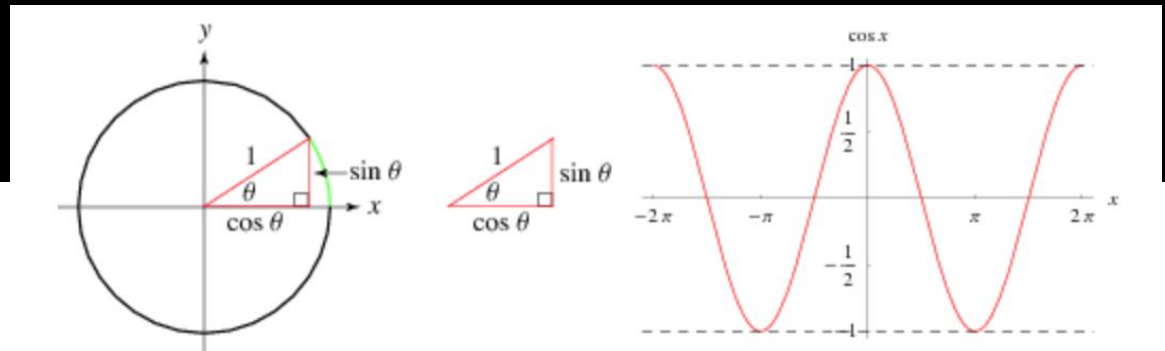
$$\cos \theta_A = r \cos \theta_Y$$

where r is the ratio between the real (A_{3D}) and projected surface area (A_{2D}) of the sample, and θ_Y is the Young contact angle.



Consequences of the Wenzel equation

- For a hydrophilic surface, i.e. when the Young contact angle $< 90^\circ$, roughness makes the surface become *more hydrophilic*
- For a hydrophobic surface, i.e. when the Young contact angle $> 90^\circ$, roughness makes the surface become *more hydrophobic*



Wetting of chemically heterogeneous surfaces, the Cassie equation

In the case of a flat, chemically heterogeneous surface, the wetting can be described by the Cassie approach. For a two component system, the wetting is described by

$$\cos \theta_A = f_1 \cos \theta_1 + f_2 \cos \theta_2$$

where f_1 and f_2 are the area fractions of material 1 and 2 ($f_1 + f_2 = 1$), and θ_1 and θ_2 the contact angles of pure materials 1 and 2, respectively.



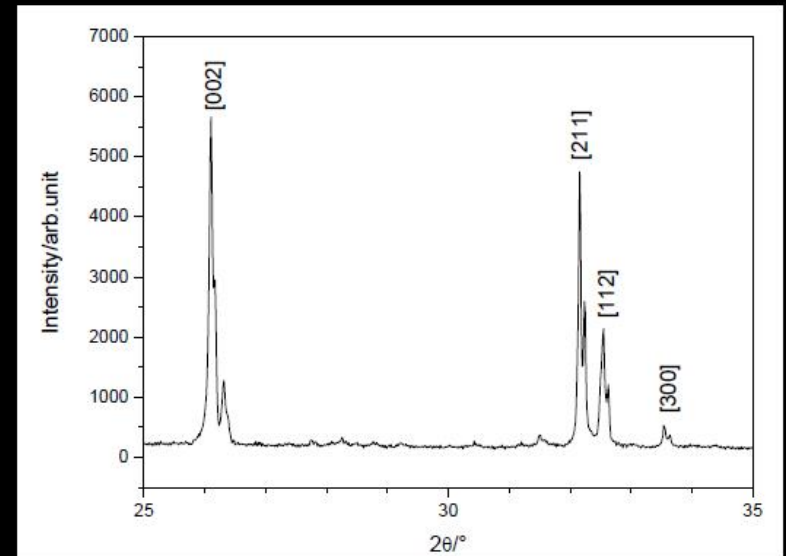
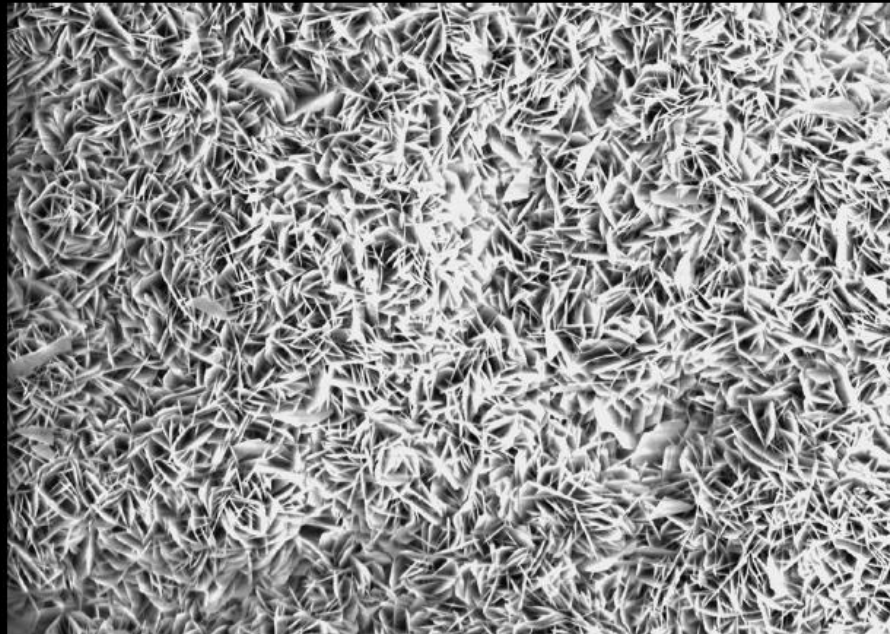
Which type of contact angles should be used in the modeling when trying to assess the chemical composition of the surface?

- Many researchers support the idea of using the advancing contact angle as the best measure, while others use the static contact angle
- The receding contact angle is less frequently used

Practical examples

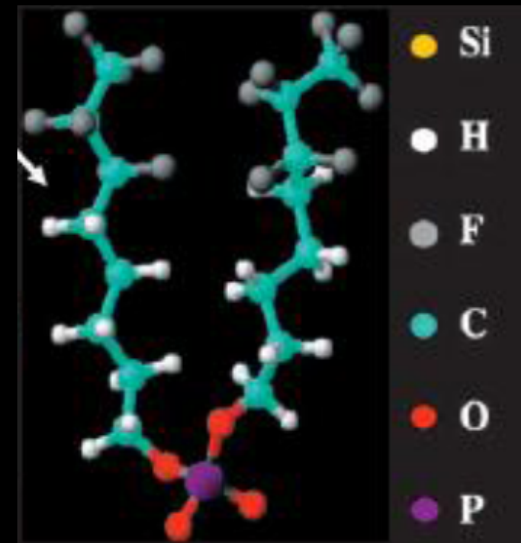
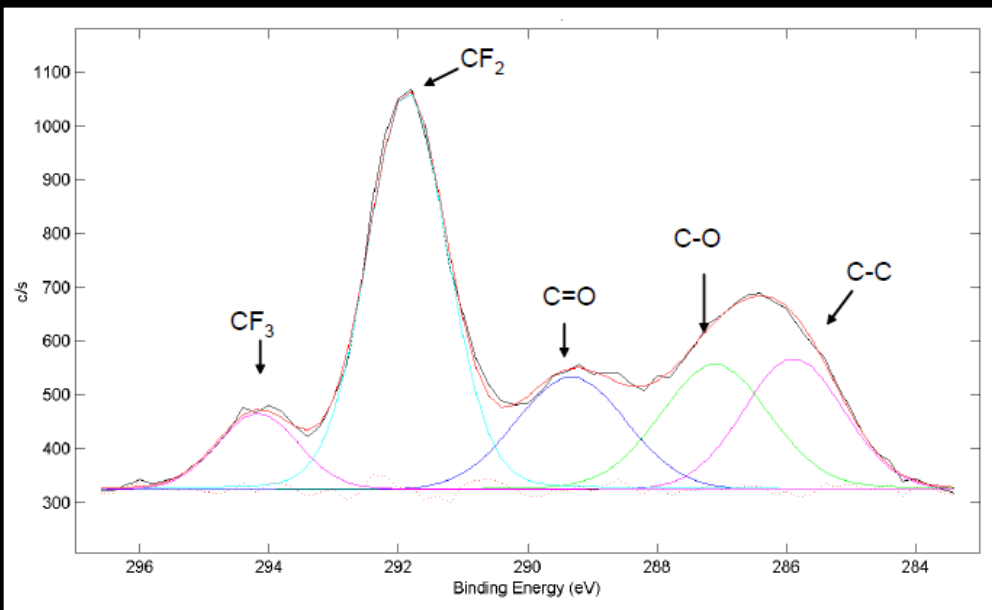
1. Superhydrophobic films

Superhydrophobic surfaces through biomineralization (生物矿化)



- The open porous structure of nucleated and grown calcium phosphate, here predominantly octacalciumphosphate, is an ideal candidate for a superhydrophobic surface, as it has a lot of open pores that can entrap air
- Surface functionalization using organophosphates is possible

Confirmation of successful zonyl surface functionalization by XPS



- XPS shows clearly that zonyl is attached to the surface
- From the CF_3 to CF_2 ratios we can estimate that the mean number of carbon atoms in the chain being either CF_3 groups or CF_2 groups is 6.5

Järn et al., Langmuir, 2008

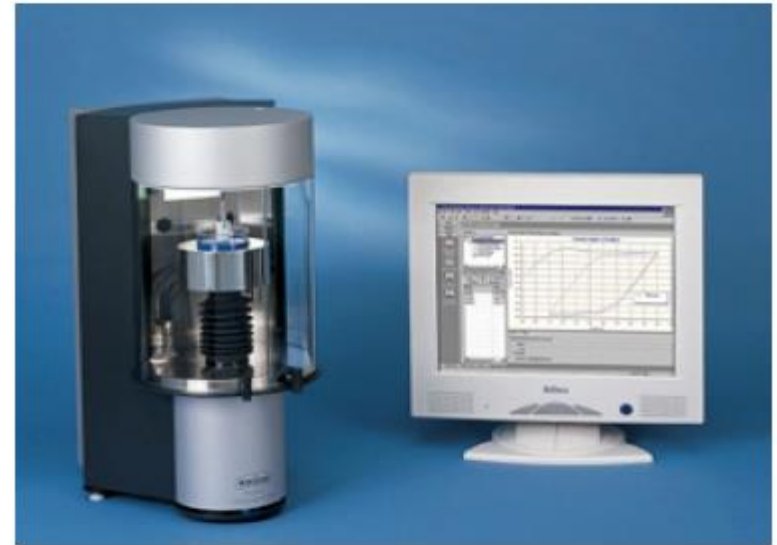
Intermediate Conclusions

- Regioselective functionalization can readily be achieved for hybrid films
- Contact angle measurements are sensitive means for determining the success of the regiofunctionalization (区域功能化)
- Contact angle measurements are NOT sensitive to the wetting properties of the surface which is covered by the droplet
- For nanopatterned films contact angle data can still be described reasonably well by determination of the relative area fractions, as the heterogeneity length scale is much smaller than the size of the droplet
- In the described cases, both the static and the advanced contact angle works reasonably well

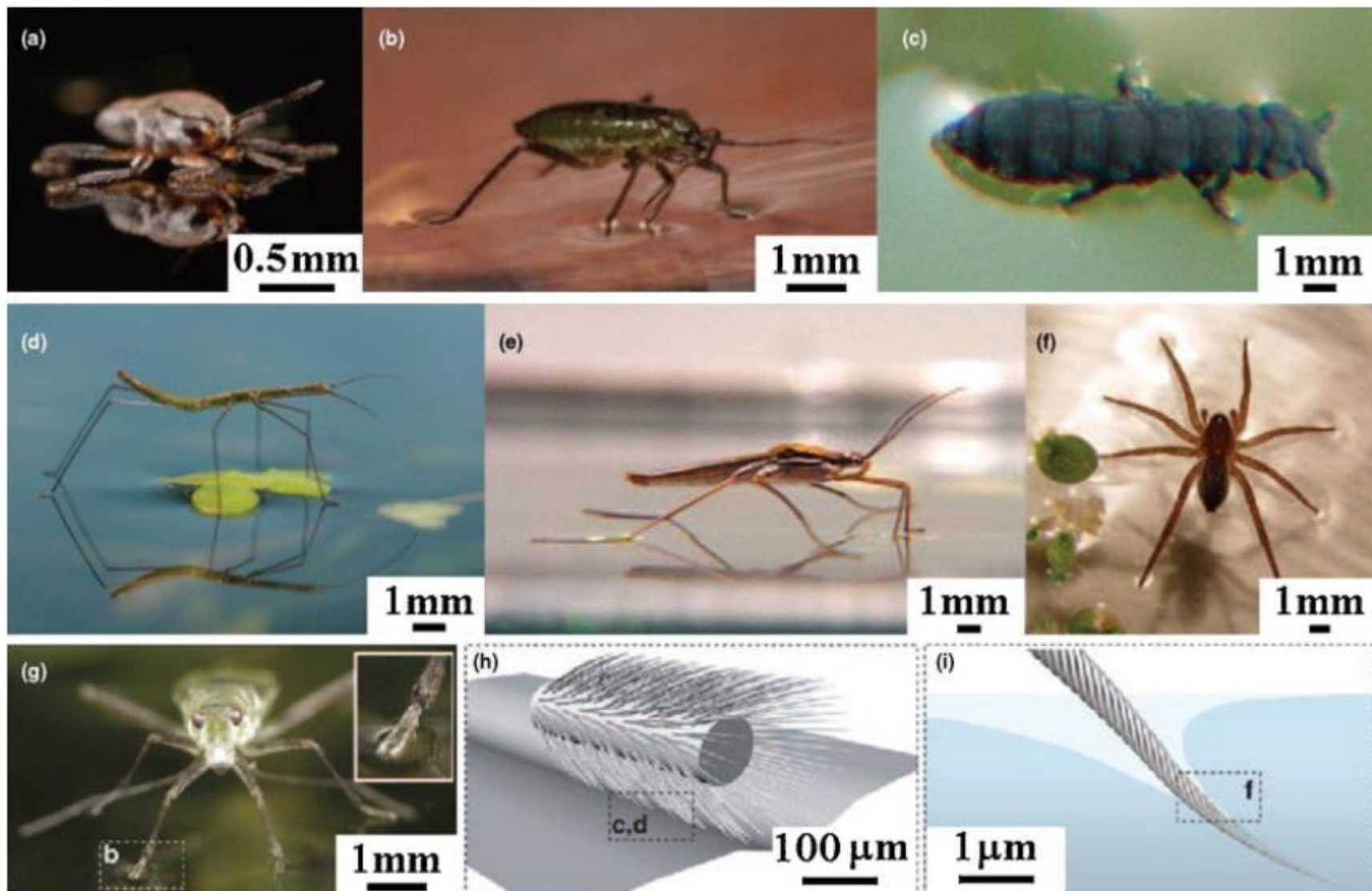
Contact angle measurement in practice



**Contact Angle Measuring
Instrument DSA100**



**Processor Tensiometer
K100**



[Download high-res image \(636KB\)](#)

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Follow up with video about contact angle measurements (see next)